

1. Model Performance

MAE (Mean Absolute Error): ~ 0.53

Interpretation: On average, our model's price prediction is off by about \$53,000 (since the target is in units of \$100,000).

RMSE (Root Mean Squared Error): ~ 0.74

Interpretation: This metric penalizes large errors more heavily. A score of 0.74 indicates reasonable performance for a simple linear model, though there is room for improvement.

2. Visual Analysis (Scatter Plot)

The plot shows a positive correlation between Actual and Predicted prices (points generally follow the diagonal line).

Observation: The model struggles to predict extremely high prices (above 5.0), where it tends to under-predict.

3. Feature Interpretation (Coefficients) The coefficients tell us how much the price changes when a feature increases by 1 unit:

Positive Impact: Features like AveBedrms (Average Bedrooms) and MedInc (Median Income) have positive coefficients. As income rises, house prices significantly rise.

Negative Impact: AveRooms has a negative coefficient in this specific multivariate context, likely due to high correlation with other features (multicollinearity).